

JINWOO SUH

Undergraduate, School of Mechanical Engineering, Yonsei University
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EDUCATION

Yonsei University

B.S. in Mechanical Engineering
3rd year

Seoul, South Korea

Feb. 2023 – Present

GPA: 3.96 / 4.30

RESEARCH INTERESTS

Robotics and Robot Learning, with a current focus on locomotion and manipulation; actively exploring other areas of the field to further narrow down my research direction.

RELEVANT COURSEWORK

Mechanism Design · Dynamics · Mechanical Vibration · SW Programming · Data Structures · Linear Algebra · Probability & Statistics · Engineering Mathematics · Computer-Aided Mechanical Design

RESEARCH EXPERIENCE

Machine Learning and Control Systems Laboratory (MLCS), Yonsei University Jun. 2026 – Sep. 2026

Undergraduate Research Intern

Advisor: Prof. Jongeun Choi

- Conducted an individual research project improving Adversarial Motion Priors (AMP), a representative locomotion policy, using a generative model (CVAE).
- Contributed to *CoHuB: A Collaborative Humanoid Loco-Manipulation Benchmark* by setting up simulation scenes and environments and implementing benchmark task configurations.
- Participated in a weekly paper review study.
- Co-organized a lecture study with fellow undergraduate interns on Deep Generative Models (Stanford CS236).

PUBLICATIONS

JOSE: Joint Only State Estimation for Whole-Body Control without an Inertial Unit

Hyeonjung Kim, Sunghyun Park, Hyungjoon Jeon, Geunrip Park, **Jinwoo Suh**

Under Review, IROS 2026 First Workshop on Sim2Real and Classical Control: From Rigorous Theory to Data-Driven Robotics

CoHuB: A Collaborative Humanoid Loco-Manipulation Benchmark

Hyunjin Park, Jebeom Chae, Minwoo Park, Soochul Yoo, **Jinwoo Suh**, Sunghyun Park, Sarmad Idrees, Jongeun Choi

Under Review, 8th Robot Learning Workshop, NeurIPS 2026

PROJECTS

Ender Buster 2 — Minecraft Combat Agent

May 2026

YAI (Yonsei Artificial Intelligence) | Team of 5 | 2nd Place, YAICON 8th

- Developed a Minecraft agent specialized in monster combat using the MineFlayer framework; referenced related work (MineDojo, Voyager, Odyssey, Jarvis-VLA) to distinguish combat-focused from planning-focused approaches.
- **Role: Testbench & Evaluation** — Built the framework used to evaluate agent performance. The resulting agent efficiently handled combat against multiple simultaneous monsters.

DEXTRA — Humanoid Sim-to-Real Walking

Nov. 2025

YAI (Yonsei Artificial Intelligence) | Team of 5 | 2nd Place, YAICON 7th

- Participated in a team project on natural humanoid locomotion via Sim-to-Real transfer using the DEXTRA humanoid platform; benchmarked DeepMimic, PPO, and AMP-based reward designs, exploring an AMP + LSTM penalty approach to balance motion naturalness and sim-to-real robustness.

ACTIVITIES

Yonsei Drone

Sep. 2024 – Sep. 2025

- ROS2 & Gazebo simulation: implemented a square-path trajectory for a quadcopter and a figure-8 trajectory for a fixed-wing aircraft using PX4-Autopilot.
- 23rd AAM Tech Challenge (Korea Robot Aircraft Competition) — contributed code debugging, flight-path waypoint construction, and YOLO data collection for object detection; Co-1st Prize (Regular Division), Sep. 2025.

YAI (Yonsei Artificial Intelligence)

Jul. 2025 – Present

- Studied deep learning fundamentals.
- Paper review study on foundational Computer Vision papers.
- Paper review study on foundational Reinforcement Learning papers.

Roboin

Mar. 2026 – Present

- Studied Arduino and Raspberry Pi.

AWARDS

Academic Honors, Yonsei University (Spring 2026)

Aug. 2026

Co-1st Prize (Regular Division), 23rd AAM Tech Challenge

Sep. 2025

Korea Robot Aircraft Competition, hosted by Korea AeroSpace Administration — awarded to Yonsei Drone

SKILLS AND LANGUAGES

Programming / Tools: Python, PyTorch, ROS2, Linux, CAD (Creo)

Languages: Korean (Native), English (TEPS 380)